

Structural Segmentation with Transformers: SpecTNT for Functional Music Structure Analysis – Extension 2

Fharook Shaik , Douglas W. Cunningham, Prashant Varadarajan

Department of Graphical Systems, Univ. BTU Cottbus-Senftenberg, Cottbus, Germany

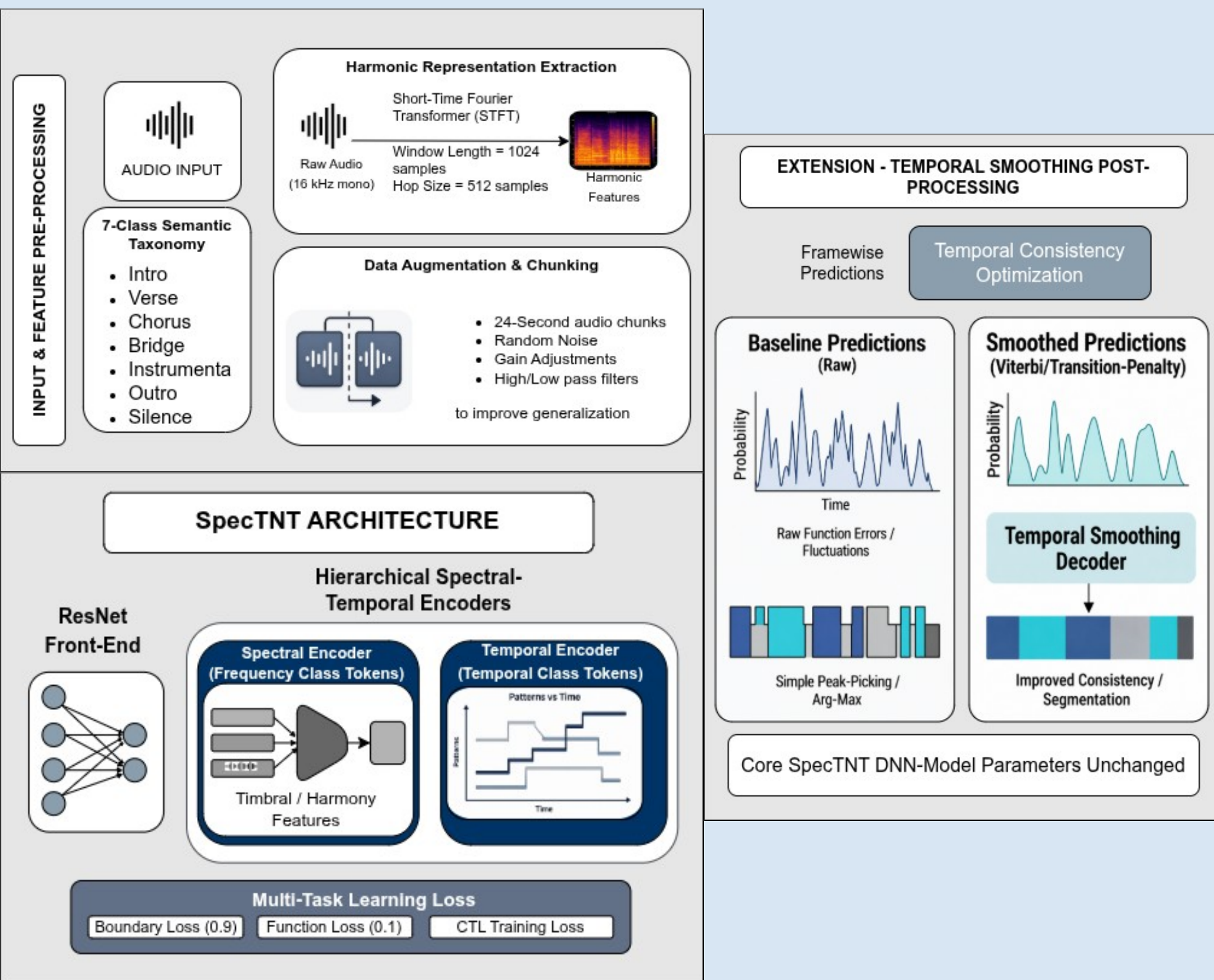
1. INTRODUCTION

Music Structure Analysis (MSA) traditionally divides songs into segments with abstract labels (A/B/C) or focuses only on chorus/boundary detection. Explicit semantic labelling (e.g., “verse”, “chorus”, “bridge”) enables powerful applications such as structure-based navigation, automated remixing, live synchronous, and real-time analysis of song fragments (e.g., TikTok Clips).

Deep-learning approaches like SpecTNT (Spectral-Temporal Transformer in Transformer) achieve strong results by predicting functional labels directly from audio [1]. However, frame-wise predictions often suffer from fragmentation and temporal inconsistency. This project reimplements the SpecTNT baseline and extends it with a lightweight temporal smoothing decoder to improve section coherence without altering the model architecture.

3. METHODOLOGY

- **Dataset:** *Harmonix Set* [2] – 912 western pop songs with functional segment annotations
- **Baseline:** *Reimplementation of SpecTNT model* [1].
- **Extension:** *SpecTNT + Temporal Smoothing Decoder (Viterbi algorithm or transition-penalty smoothing) as post processing step.*
- **Evaluation Metrics:** *Hit Rate (HR.5F), Frame-wise Accuracy (ACC), Pair-Wise F-measure (PWF), Normalized Entropy Score (Sf), Macro F1 (7 classes), Chorus Hit Rate (CHR.5F).*



5. CONCLUSION

- **Chorus detection is a relative strength:** CF1=0.68 – Model learns to distinguish chorus from non-chorus sections effectively even with limited data.
- **Base model outperforms CTL** – CTL requires more diverse training data to generalize
- **Viterbi smoothing degrades performance** – Model’s frame level predictions are more accurate than what the transition model enforces. Data-driven transition matrix doesn’t align with the model’s learned patterns.
- **Smoothing cannot fix boundary errors** – it only relabels frames within segments, leaving over-segmentation unchanged.
- **Future Scope:** Train on multiple datasets to improve CTL generalization. Adjust peak-picking to reduce over-segmentation. Address class imbalance with focal loss or oversampling.

2. RESEARCH QUESTION

Following Wang et al. (2022) [1], this project performs functional structure analysis using a 7-class taxonomy: *intro, verse, chorus, bridge, instrumental, outro, silence*.

How does applying a temporal smoothing decoder (Viterbi algorithm or a transition-penalty smoothing) as post-processing on top of SpecTNT’s frame-wise predictions affect overall accuracy and temporal consistency of the predicted musical sections?

Hypothesis: Because the sequence of sections in popular music follows regular patterns, explicitly smoothing the frame-wise probabilities during post-processing will significantly improve structural segmentation metrics (HR.5F, PWF, Sf) and macro F1 scores without requiring modifications to the underlying neural network architecture

4. RESULTS

Metric	SpecTNT (24s, No CTL)			SpecTNT (24s, CTL)		
	Baseline	Baseline + Viterbi Smoothing	Δ	Baseline	Baseline + Viterbi Smoothing	Δ
HR.5F	0.2767	0.2767	0	0.2748	0.2748	0
ACC	0.4932	0.4896	-0.0036	0.4577	0.4580	0.0004
PWF	0.5544	0.5491	-0.0053	0.5344	0.5212	-0.0131
Sf	0.2325	0.2082	-0.0243	0.1970	0.1674	-0.0295
CHR.5F	0.1576	0.1518	-0.0058	0.1604	0.1560	-0.0044
CF1	0.6805	0.6795	-0.0009	0.6574	0.6515	-0.0059
Macro F1	0.1497	0.1420	-0.0077	0.1422	0.1352	-0.0070

Table 1: Effect of Viterbi smoothing on SpecTNT (24s) with and without CTL.

